



Manual

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# Overview

Optimsolution is a C++ optimization framework designed for repeatable, high-throughput experimentation with population-based metaheuristics and numerical optimizers on benchmark and application-driven objective functions. Its primary objective is to standardize execution (initialization, stopping rules, logging) so that results are comparable and fully reproducible across methods, problems, and runs.

The project supports two primary usage modes: a command-line interface (CLI) for batch execution and automation, and a graphical user interface (GUI) for interactive method/problem selection, live log inspection, and convergence visualization via CSV output.

## What this manual covers

This manual provides step-by-step installation and build instructions for Windows and Linux (Debug and Release), covers both CLI and GUI usage, explains the configuration model (optimsolution.cfg, the [global] section, and runtime overrides), and describes the expected output artifacts (run summaries, CSV files, and convergence traces).

## Core capabilities

- Unified interface for running methods on problems, with consistent logging and a standardized end-of-run summary.
- Explicit termination budgets (function evaluations and/or iterations), suitable for benchmarking protocols.
- Reproducibility support through independent runs and consistent initialization/stopping conventions.
- Configuration via optimsolution.cfg, including shared defaults under [global] and per-run overrides.
- CSV export and convergence recording to enable plotting and cross-run comparisons.
- GUI with separate method/problem dropdowns, Run/Stop process control, single-line log entries, and per-run output tabs.

## Terminology and key concepts

The following terms are used consistently throughout this manual. Maintaining strict terminology is important for correct interpretation of settings and output files.

- Method: The algorithm/optimizer being executed (e.g., DE variants, CMA-ES, etc.).
- Problem: The objective function/benchmark with defined bounds and dimension.
- Run: One independent execution of a method on a problem under a specific budget and configuration.
- Budget: The computational limit (evaluations/iterations) that determines when a run terminates.
- CSV / Convergence: Output files capturing best-so-far progress during the run.

- Run Summary: The standardized end-of-run report (e.g., best/mean, timing, success criteria when defined).

## What to expect next

The next chapters provide concrete build commands, practical guidance for separating CLI and GUI settings, and execution examples. Instructions are written to support clean builds and reproducible runs from any working directory without ambiguity.

# System Requirements

optimsolution is a C++ project built with CMake. It can be used as a console application (CLI) and, optionally, as a Qt6-based graphical application (GUI). Requirements differ depending on whether you build only the CLI or both CLI and GUI.

## Supported Platforms

- Windows 10/11 (x64) with a Visual Studio / MSVC toolchain.
- Linux (x64) with GCC or Clang.

The codebase targets a 64-bit environment. Exact package names and availability on Linux depend on the distribution and the selected toolchain.

## Dependencies

### Required (CLI and GUI)

- CMake.
- A C++ compiler with C++17 support

### Optional (GUI only)

- Qt6 (Core, Widgets, and any modules required by the GUI target).
- A Qt kit that matches the compiler toolchain (e.g., an MSVC kit on Windows).

On Windows, the most common build failures are caused by a mismatch between the compiler and the Qt build (e.g., MSVC vs MinGW). On Linux, a frequent cause is an incorrect or undiscoverable Qt prefix path.

# Windows Installation and Build

Windows installer available (fastest path): download and run Optimsolution.msi from: <http://dit.uoi.gr/files/Optimsolution.msi>

This chapter describes the Windows 10/11 build workflow using MSVC and CMake. Instructions are organized by CLI and GUI, and by Debug and Release configurations. The GUI requires Qt6 and a correct Qt prefix path.

## Prerequisites

- Visual Studio 2022 with the “Desktop development with C++” workload installed.
- CMake available on PATH (or via Visual Studio).
- A Qt6 MSVC kit (e.g., C:\Qt\6.10.1\msvc2022\_64) to build the GUI.

It is recommended to run commands from “Developer PowerShell for VS 2022” so the MSVC environment is correctly initialized.

## Console (CLI) – Debug

Clean configure/build (out-of-source):

```
Remove-Item -Recurse -Force .\build -ErrorAction SilentlyContinue
cmake -S . -B build -G "Visual Studio 17 2022" -A x64 `
  "-DCMAKE_PROJECT_INCLUDE:FILEPATH=$PWD/cmake/optimsolution_gui.cmake" `
  "-DCMAKE_PREFIX_PATH=C:\Qt\6.10.1\msvc2022_64"
cmake --build build --config Debug --target optimsolution
.\build\Debug\optimsolution.exe --help
```

With a multi-config generator (Visual Studio), Debug/Release selection happens at build time via --config. CMAKE\_BUILD\_TYPE is not used in this scenario.

## Console (CLI) – Release

```
cmake --build build --config Release --target optimsolution
.\build\Release\optimsolution.exe --help
```

You may reuse the same build directory for Debug and Release. If you change generator/toolchain, perform a clean build.

## Graphical User Interface (GUI) – Debug

The GUI target is enabled through Qt6. Two common integration patterns are used: (a) specifying CMAKE\_PREFIX\_PATH at configure time, or (b) using CMAKE\_PROJECT\_INCLUDE to load cmake/optimsolution\_gui.cmake.

### Pattern A: CMAKE\_PREFIX\_PATH

```
Remove-Item -Recurse -Force .\build -ErrorAction SilentlyContinue
cmake -S . -B build -G "Visual Studio 17 2022" -A x64 `
  "-DCMAKE_PROJECT_INCLUDE:FILEPATH=$PWD/cmake/optimsolution_gui.cmake" `
  "-DCMAKE_PREFIX_PATH=C:\Qt\6.10.1\msvc2022_64"
```



```
cmake --build build --config Debug --target optimsolution_gui
& "C:\Qt\6.10.1\msvc2022_64\bin\windeployqt.exe" `
--no-translations --compiler-runtime `
".\build\Debug\optimsolution_gui.exe"
.\build\Debug\optimsolution_gui.exe
```

## Pattern B: CMAKE\_PROJECT\_INCLUDE (GUI include)

If the repository is structured to enable GUI integration via a separate CMake include, pass the include file at configure time. This keeps GUI-related CMake logic encapsulated.

```
Remove-Item -Recurse -Force .\build -ErrorAction SilentlyContinue
cmake -S . -B build -G "Visual Studio 17 2022" -A x64 `
"-DCMAKE_PROJECT_INCLUDE:FILEPATH=$PWD/cmake/optimsolution_gui.cmake" `
"-DCMAKE_PREFIX_PATH=C:\Qt\6.10.1\msvc2022_64"
cmake --build build --config Debug --target optimsolution_gui
.\build\Debug\optimsolution_gui.exe
```

## Graphical User Interface (GUI) – Release

```
cmake --build build --config Release --target optimsolution_gui
.\build\Release\optimsolution_gui.exe
```

## Common Issues and Fixes (MSVC/Qt)

### Qt kit mismatch (MSVC vs MinGW)

Ensure the Qt installation used by CMake is built for MSVC (e.g., msvc2022\_64). Otherwise, link errors or binary incompatibilities are expected.

### CMake cannot find Qt

- Verify -DCMAKE\_PREFIX\_PATH points to the correct Qt prefix.
- Alternatively, set Qt\_DIR to a path containing the Qt6\*Config.cmake files.
- Perform a clean configure when changing Qt paths.

### ‘identifier not found’ errors for Qt classes

These errors typically indicate missing headers or the use of API calls that do not match the selected widget class.

- If you use QPlainTextEdit, ensure <QPlainTextEdit> is included.
- If you use QTextEdit and need to limit log lines, use textEdit->document()->setMaximumBlockCount(N) rather than textEdit->setMaximumBlockCount(...), which is not available on QTextEdit.

## **GUI launches but Qt DLLs are missing**

Debug/Release builds require the corresponding Qt runtime DLLs. If you run outside Qt Creator, use windeployqt or ensure the Qt bin directories are on PATH. Proper deployment is typically handled via packaging when distributing binaries.

## **General recommendation when changing build settings**

- If you change generator, toolchain, or Qt prefix, delete build/ and reconfigure from scratch.
- Use separate build directories if you need multiple Qt kits or parallel configurations.

# Linux Installation and Build

This chapter describes the Linux build workflow using CMake and GCC/Clang. The build is typically single-config (CMAKE\_BUILD\_TYPE), and out-of-source builds into a dedicated build directory are recommended.

## Prerequisites

On Debian/Ubuntu-based distributions, install Qt6 development packages for the GUI target:

```
sudo apt update
```

```
sudo apt install qt6-base-dev qt6-base-dev-tools qt6-tools-dev qt6-tools-dev-tools
```

- CMake available on PATH.
- GCC or Clang with C++17 support.
- For the GUI: Qt6 (Core/Widgets) installed and discoverable by CMake.

Package names vary by distribution. If you rely on system packages, ensure the corresponding development packages are installed for Qt6 and any other required dependencies.

## Console (CLI) – Debug

With single-config generators, Debug/Release is set at configure time using CMAKE\_BUILD\_TYPE.

```
rm -rf build
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Debug -
DCMAKE_PROJECT_INCLUDE=cmake/optimsolution_gui.cmake
cmake --build build --target optimsolution
./build/optimsolution --help
```

If your target name or executable path differs, use the build output or list the build/ directory to locate the produced binary.

## Console (CLI) – Release

```
rm -rf build
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Release -
DCMAKE_PROJECT_INCLUDE=cmake/optimsolution_gui.cmake
cmake --build build --target optimsolution
./build/optimsolution --help
```

For benchmarking, a Release build is recommended. Debug builds are primarily used for development and debugging.

## Graphical User Interface (GUI)

The GUI requires Qt6 development packages. The README enables the GUI target via CMAKE\_PROJECT\_INCLUDE=cmake/optimsolution\_gui.cmake.

## Configure (GUI enabled)

```
rm -rf build
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=Debug -
DCMAKE_PROJECT_INCLUDE=cmake/optimsolution_gui.cmake
cmake --build build --target optimsolution_gui
./build/optimsolution_gui
```

For detailed GUI guidance (layout, tabs, logging, and convergence plotting), refer to [optimsolution\\_manual.pdf](#). For a visual reference of the GUI, use the figure [optimsolution\\_gui.png](#).

## Common Issues and Fixes (toolchain/Qt)

### CMake cannot find Qt6

- Confirm Qt6 development packages are installed (distribution-specific names).
- If using an installer-based Qt, ensure CMAKE\_PREFIX\_PATH points to the correct Qt prefix.
- Delete build/ and reconfigure when switching Qt paths or toolchains.

### Link errors or missing shared libraries at runtime

If the executable builds but fails to run due to missing shared libraries, use `ldd` to inspect dependencies and ensure the runtime environment (`LD_LIBRARY_PATH` / `rpath`) is correctly configured.

### Switching compilers

If you switch from GCC to Clang (or vice versa), prefer a clean build in a new build directory to avoid mixing artifacts.

## CLI Usage

The optimsolution CLI is designed for batch execution and automation. It enables running a selected method on a selected problem under an explicit budget with consistent stopping behavior, producing standardized logs and an end-of-run summary.

### Basic syntax

Always use the built-in help to confirm the available syntax in your build.

```
optimsolution --help
```

### Selecting method and problem

Note: The README demonstrates method/problem selection as positional arguments (e.g., "optimsolution arq tersoffc"). Use "optimsolution --help" to confirm any optional arguments supported by your build.

Typical workflow: set method and problem (using their short names).

```
optimsolution <METHOD> <PROBLEM>
```

```
optimsolution <METHOD> <PROBLEM>
```

For fixed-dimension problems, the CLI may ignore --dim or provide an alternative mode.

### CSV and convergence output (csv\_enable/csv\_convergence)

- csv\_enable: enables/disables CSV creation.
- csv\_convergence: enables convergence recording (best-so-far).

The GUI uses these CSV files to render convergence plots.

### Run summary and relationship to configuration

Each run ends with a standardized run summary. Most parameters are read from optimsolution.cfg

# Settings and Configuration Files

optimsolution relies on explicit configuration to support reproducibility and standardized execution. The primary configuration source is `optimsolution.cfg` located at the repository root. This manual follows a layered approach: shared defaults live in `[global]`, while specialization is applied through dedicated sections and/or runtime overrides.

## `optimsolution.cfg`

`optimsolution.cfg` is an INI-style configuration file with sections. It typically includes general runtime settings, method/problem parameters (where applicable), and output controls (logs/CSV).

Key defaults (from README):

```
[global]
population = 200
max_iters = 500000
max_evals = 150000
seed_base = 4242
runs      = 30
success_tol = 1e-8
local_rate = 0.00
local_method = lbfgs
end_local_refine = 0
end_local_method = lbfgs ; or bfgs / nm / gd
summary_enable = 1

[stop]
rule = maxevals      ; bss|wss|tss|boss|srs|irs|doublebox|maxevals|all|none
eps  = 1e-10

[init]
type = uniform      ; uniform | normal | cauchy | laplace | lognormal | exponential | beta |
levy | lhs | halton | oppositional
```

Example method overrides (ARQ):

```
[arq]
population = 100
pbest_frac = 0.12
F          = 0.1
CR         = 0.9
archive_rate = 1.5
batch_frac  = 0.5
local_rate  = 0.00
local_method = lbfgs
```

Exact key/section names depend on the implementation. The example communicates the intent: shared defaults in [global], specialized values per method/problem when needed, and a separate section for GUI state.

## [global] and runtime overrides

The [global] section acts as a default layer. More specific sections (e.g., method-specific) and runtime overrides take precedence over global values. This allows you to keep a stable baseline while changing only what is required for a particular run.

- Layer 1: [global] defaults.
- Layer 2: section overrides (e.g., method/problem sections).
- Layer 3: runtime overrides (parameters provided on the command line or set in the GUI before starting a run).

Runtime overrides are useful for parameter sensitivity studies and quick experiments without permanently editing the cfg. A critical requirement is that overrides must be recorded alongside the run to ensure full reproducibility.

## Per-run temporary configuration snapshot

To ensure each run is fully documented, optimsolution creates a temporary snapshot of the effective configuration used for that run (global + overrides) and stores it together with the run outputs. This snapshot allows you to reproduce a run exactly even if the base optimsolution.cfg changes later.

Recommended practice:

- Store the snapshot cfg inside the run output directory (next to CSV and summary files).
- Log the snapshot path and the applied overrides in the run log.
- Use a unique run id or timestamp to avoid accidental overwrites.

## Separating CLI settings vs GUI settings

The CLI and GUI share the same execution core settings (budgets, CSV, output paths, etc.). However, the GUI also needs user-experience state (e.g., last selected method/problem, layout preferences, plot visibility). To avoid confusion and to preserve experimental integrity, these settings should be clearly separated.

- CLI settings: anything that can affect numerical outcomes (e.g., budgets, seed, method parameters, CSV settings).
- GUI settings: UI state and presentation only (e.g., last selected method/problem, window geometry, plot options).

In practice, optimsolution.cfg may include a dedicated GUI section (e.g., [gui]) and/or the GUI may use a separate settings file. In all cases, the per-run snapshot should include only execution-relevant settings, not pure UI state, so the experimental footprint remains clean.

## Minimal configuration for experiments

For typical benchmarking runs, keep the following in [global] and vary only what is strictly required per method:

```
[global]
max_evals      = 200000
csv_enable     = 1
csv_convergence = 1
output_root    = outputs
```

Then, define method-specific parameters in method sections so it is explicit what changes per method. Use runtime overrides only for ad-hoc changes and ensure they are captured in the per-run snapshot.



# GUI Usage

The optimsolution GUI provides interactive method/problem execution with a focus on a clear workflow, live log inspection, and fast convergence visualization via CSV output. For a visual reference of the layout, see `optimsolution_gui.png`. For detailed GUI instructions, refer to `optimsolution_manual.pdf`.

## Window layout and workflow

The standard workflow is: select a Method, select a Problem, adjust parameters/budgets where applicable, then start execution with Run. Each execution produces log output and opens an output tab named as method+problem.

## Method/Problem selection

Selection is performed via two separate dropdowns. The method dropdown displays entries as “Full name (short)” to remain both readable and compact.

- Method dropdown: “Full name (short)”.
- Problem dropdown: a recognizable short/full name depending on the project.

## Dimension handling (fixed vs variable)

For fixed-dimension problems, the GUI does not expose a dimension control and does not apply dimension auto-retry. For variable-dimension problems, a dimension field/control is shown when required.

- Fixed-dimension: hide the dimension control and avoid auto-retry.
- Variable-dimension: show the dimension control and pass it as a run parameter.

## Run/Stop and process control

The execution button acts as a Run/Stop toggle. While running, Stop terminates the current process in a controlled manner to avoid orphan processes.

- Run: start an execution using the current configuration.
- Stop: terminate the current execution and return to idle state.

## Log view

The log is presented as a sequence of single-line entries without blank lines. This improves readability and makes copying checkpoints (e.g., best values) straightforward without formatting noise.

## Output tabs

Outputs are organized into multiple tabs. Each tab corresponds to a run (or logical execution) and is labeled using method+problem. Tabs are closable to keep the workspace clean.

## Convergence plot from CSV

The convergence plot is built from CSV outputs (`csv_enable/csv_convergence`). The GUI reads the CSV files and renders best-so-far curves. The plot is designed to support comparing multiple problems under the same method within a single chart.

### Axes and labels

- X-axis title: “Iterations”.
- Y-axis title: “Best solution”

### Legend and overlay

The legend includes the problem short name and the dimension. The GUI supports overlaying multiple problems for the same method to compare curves in a single figure.

### Export PNG

The plot can be exported as a 300 DPI PNG for reports and papers. Export should preserve axis titles and a readable legend.

## Colored RUN SUMMARY styling

At the end of execution, the GUI renders a color-styled RUN SUMMARY to make key fields (e.g., best value, success, timing) immediately visible. This complements the raw log output and enables fast result inspection.

# Examples

This chapter provides practical examples for running optimsolution via the CLI and the GUI, along with a minimal configuration baseline. Examples are intentionally template-style so they can be adapted to the actual method/problem names available in your build.

## CLI run example

Run a single execution

optimsolution <METHOD> <PROBLEM>

```
Windows PowerShell
[Run 30 | iter 2491 | evals 149794] best_f = -32.013207087264
[Run 30 | iter 2492 | evals 149854] best_f = -32.013207087264
[Run 30 | iter 2493 | evals 149914] best_f = -32.013207087264
[Run 30 | iter 2494 | evals 149974] best_f = -32.013207087264
[Run 30 | iter 2495 | evals 150000] best_f = -32.013207087264

===== RUN SUMMARY =====

----- GENERAL -----
Method: arq
Method full name: ARQ: Adaptive RTR with Quarantine
Problem: tersoffc (Dim=24)

----- PROBLEM INFO -----
Problem full name: TersoffC Si(C) Cluster Potential (CEC2011)
Problem modality: multimodal
Problem separability: non-separable
Problem category: molecular modelling / cluster optimization
Bounds: non-uniform per dimension

----- EXPERIMENT CONFIG -----
Population: 100
Init distribution: uniform
Runs: 30
Max evals/run: 150000
Max iters/run: 500000
Stop rule: maxevals
Success criterion: f within 3.42574055152e-08 of best-of-runs (-34.2574055152)

----- PERFORMANCE -----
Best f (min): -34.257405515172
Best f (mean +- sd): -32.692619515222 (+- 0.672258983381)
Best f (median, Q1-Q3): median=-32.7007 [Q1=-33.2032, Q3=-32.237]
Evals to reach target: 150000 (+- 0)

----- CALLS & SUCCESS -----
Evals (mean over runs): 150000 (+- 0)
Grad calls (mean over runs): 0 (+- 0)
Grad/Func calls ratio (mean): 0
Success rate: 3.33333% (1/30)
Best run (index,f,evals): #1 (f=-34.2574, evals=150000)
Worst run (index,f,evals): #2 (f=-31.3558, evals=150000)

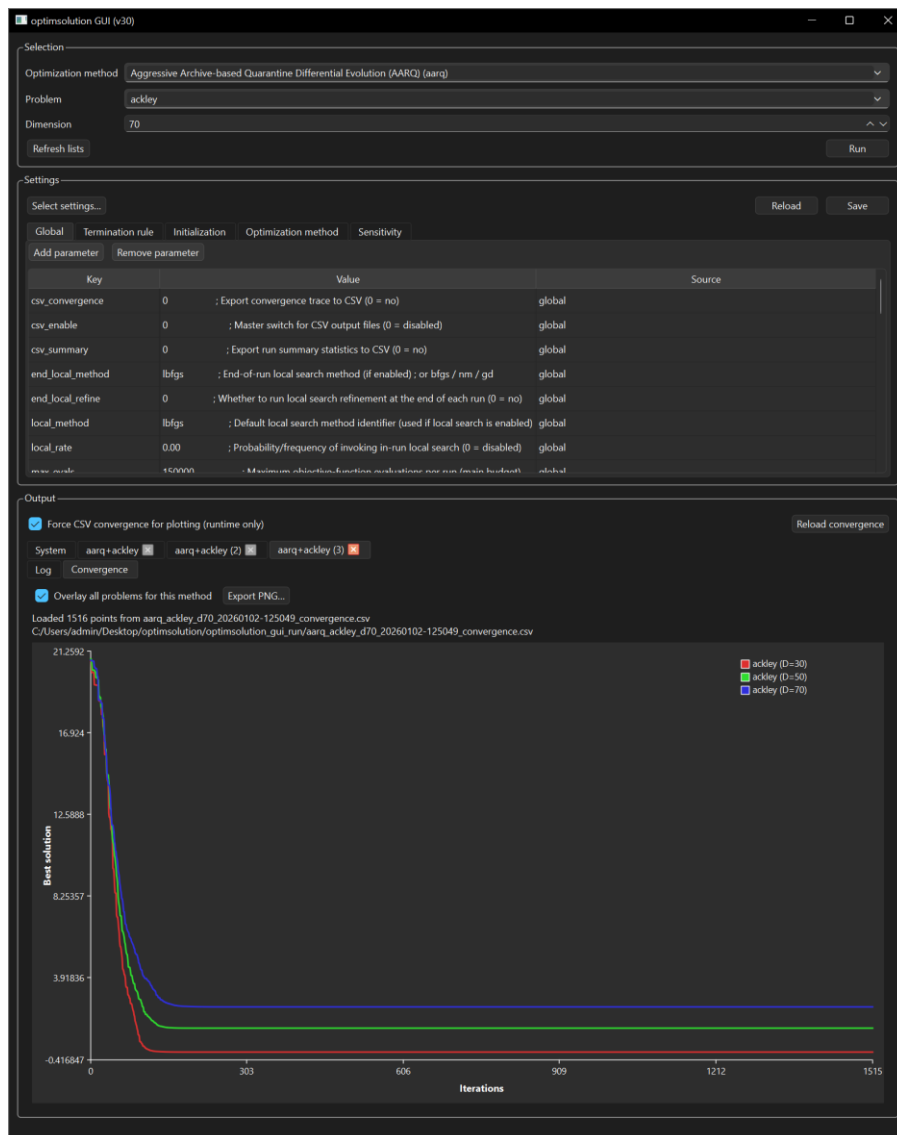
----- TIMING & MEMORY -----
Iter time (mean of means): 0.006155 s (+- 0.000175 s)
Iter time (p95 across runs): 0.007548 s (+- 0.000631 s)
Time per run (mean): 15.329133 s
Total wall time (all runs): 459.873993 s
Memory peak (mean): 5.87018 MB (+- 0.00499226 MB)

----- LOCAL SEARCH -----
Local search (in-run): none
Local search (in-end): none

----- SUMMARY HIGHLIGHTS -----
Best f (min): -34.2574
Success rate: 3.33333% (1/30)
PS C:\Users\admin\Desktop\optimsolution\build>
```

## GUI run example

- 1) Launch the GUI executable (typically under build/Debug or build/Release on Windows).
- 2) Select a Method from the dropdown (displayed as “Full name (short)”).
- 3) Select a Problem from the corresponding dropdown.
- 4) If the problem is variable-dimension, set the dimension. For fixed-dimension problems, the dimension control is hidden.
- 5) Click Run and monitor the log (single-line entries).
- 6) At completion, inspect the color-styled RUN SUMMARY and the output tab (method+problem).
- 7) If csv\_enable/csv\_convergence are enabled, view the convergence plot and export a 300 DPI PNG if needed.



## Minimal configuration example (optimsolution.cfg)

The snippet below is a minimal, practical baseline for benchmarking. Adjust `max_evals` and `output_root` to match your workflow.

```
[global]
population = 200
max_iters = 500000
max_evals = 150000
runs      = 30
seed_base = 4242
csv_enable = 1
csv_convergence = 1
output_root = outputs
[stop]
rule = maxevals
```

```
[init]
type = uniform
```

## optimsolution.cfg not found from subfolders

Symptom: when running the CLI/GUI from `build/Debug`, `build/Release`, or another subfolder, the application fails to locate `optimsolution.cfg` even though the file exists at the repository root.

Practical fixes:

- Set the run working directory to the repository root (especially when launching from an IDE).
- Implement discovery logic that walks parent directories until `optimsolution.cfg` is found.
- Support an explicit option such as `--config <path_to_cfg>` to provide an absolute configuration path.
- Log the resolved configuration path so the issue is immediately visible in logs.

Note: this manual considers it desirable that the root `cfg` can be located regardless of the working directory, either through robust discovery or an explicit `config` path.

## Qt build/link issues (Windows)

### QPlainTextEdit: 'identifier not found'

If `QPlainTextEdit` is not recognized, a missing include or missing linkage to the correct Qt modules is a typical cause.

- Add the header include: `<QPlainTextEdit>`.
- Ensure the target links against `Qt6::Widgets`.
- Perform a clean build after changing modules or the Qt prefix.

### QTextEdit: setMaximumBlockCount is not available

setMaximumBlockCount is an API of QPlainTextEdit. For QTextEdit, limit blocks via the document().

```
textEdit->document()->setMaximumBlockCount(1000);
```

Ensure <QTextEdit> and <QTextDocument> are included as required.

## Qt kit mismatch (MSVC vs MinGW)

Use a Qt build that matches the compiler toolchain used to build the project. For MSVC builds, the Qt prefix must be an MSVC kit (e.g., msvc2022\_64).

## Qt build/link issues (Linux)

On Linux, common causes are missing Qt6 development packages or CMake failing to locate Qt because the prefix path is incorrect.

- Confirm Qt6 development packages are installed (distribution-specific names).
- If using an installer-based Qt, set a correct -DCMAKE\_PREFIX\_PATH at configure time.
- Delete build/ and reconfigure when switching Qt paths.

## Missing or incorrect CSV / convergence output

Symptom: the GUI does not display a plot or shows an empty/incorrect figure even though a run completed.

Checks and fixes:

- Verify csv\_enable=1 and csv\_convergence=1 in the effective configuration (snapshot).
- Confirm the run output directory is what you expect (e.g., output\_root).
- Ensure CSV files are created and contain data (not headers only).
- If plotting relies on naming conventions, ensure filenames match the expected pattern.

If the CSV is correct but the plot remains empty, confirm the GUI is reading the same output directory that the run wrote to, and that multiple working directories are not producing divergent outputs.

## Run/Stop does not terminate execution correctly

Symptom: pressing Stop does not stop execution, or an orphaned process remains.

- Ensure the GUI holds the correct process handle and terminates the correct PID.
- Log state transitions (Run -> Stop) and the termination outcome.
- If child processes exist, ensure they are terminated as well (implementation-dependent).

## General diagnosis strategy

- First, verify paths/working directory and the effective configuration snapshot.
- Then, verify toolchain/Qt mismatches and perform clean reconfigure after changing core build parameters.

- Finally, inspect outputs (CSV, logs) and confirm the GUI/CLI point to the same output\_root.

## Help: Why 30 independent runs?

Many optimizers implemented in `optimsolution` are stochastic (e.g., evolutionary and swarm methods). In such algorithms, the observed outcome is not a single deterministic value; it is a random variable that depends on sources of randomness such as the initial population, mutation/crossover choices, and other sampling operations. For this reason, a single execution may be unrepresentative (either unusually good or unusually poor), and conclusions based on one run are generally not statistically defensible.

### Reliability (repeatability) in stochastic optimization

In measurement science, reliability refers to the consistency of a measurement procedure when repeated under identical conditions. In stochastic optimization, reliability is assessed by repeating the algorithm under the same experimental configuration (same problem, bounds, budget, and parameterization) while varying only the random seed. If the solver produces similar performance distributions across repetitions (e.g., stable mean/median and reasonable variance), then the experimental outcomes are considered repeatable and the method can be evaluated with confidence.

### Validity (accuracy with respect to the objective)

Validity refers to whether a measurement captures the “true” quantity of interest. In optimization benchmarking, validity is linked to whether the experimental protocol can credibly reflect the algorithm’s ability to approach the best achievable objective values under the given budget. A reliable protocol is a prerequisite for validity: without repeatability, it is impossible to distinguish genuine algorithmic performance from random fluctuations.

### Why “30 runs” is a practical standard

Using multiple independent runs is required to:

- estimate central tendency (mean/median) and dispersion (standard deviation / interquartile range),
- quantify success probability (success rate) under a fixed budget,
- support inferential comparisons between methods using non-parametric tests (e.g., Wilcoxon, Friedman) that assume independent samples.

Choosing 30 independent runs is widely used as a practical compromise between statistical stability and computational cost. With ~30 samples, summary statistics and empirical success rates typically become markedly more stable than with very small sample sizes, and the sample size is also sufficient for many standard non-parametric comparisons to have reasonable power under typical benchmarking budgets.

### Recommended practice in `optimsolution`

Use 30 runs with different random seeds for each (problem, dimension, method, configuration) tuple.

- Report at minimum: best-of-runs, mean  $\pm$  sd, median (Q1–Q3), and success rate.



- Keep everything else fixed (budget, bounds, initialization type, parameter settings) so that observed variability is attributable to stochasticity rather than uncontrolled changes.